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Clinical review of robotic distal pancreatectomy

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≈1 day shorter hospital stay than laparoscopic

0% positive margins (laparoscopic 2%)

≤ LDP overall cost; well below open surgery

+30 min operating time vs laparoscopic, mostly docking

01 Background & aim

- Distal pancreatectomy has moved from open (ODP) to laparoscopic (LDP) and robotic (RDP) surgery.
- LDP is limited by dexterity and visualisation.
- Whether RDP's technical gains improve outcomes is still debated.

Aim Compare RDP with ODP and LDP: technique, outcomes, cost and challenges.

02 Why robotic?

3D magnified vision

Tremor filtration

Motion scaling

Wristed instruments

Platforms: almost all evidence is from da Vinci; Versius and hinotori™ have only early case reports.

03 Operative technique

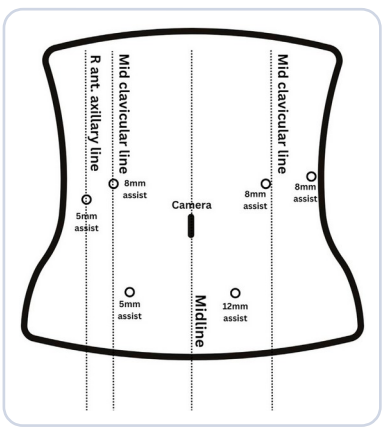


Fig. 1 Port placement

1 **Position** Supine or split legs; steep reverse Trendelenburg; dock

2 **Expose** Enter the lesser sac

3 **Control** Splenic artery and splenic vein

4 **Divide** Pancreas with a macroscopic margin

5 **Resect** En bloc retroperitoneal dissection ± spleen

Spleen preserving: splenic vessel preservation or Warshaw technique

04 Clinical outcomes of RDP

Outcome	RDP vs open	RDP vs laparoscopic
Blood loss	▼ **Lower**	▼ **Lower (planned spleen preservation)**
Complications	▼ **Fewer (fistula, respiratory, wound)**	= Similar overall, major and fistula
Hospital stay	▼ **Shorter**	▼ **≈1 day shorter**
Conversion to open	Not applicable	▼ **Fewer**
Spleen preservation	▲ **Higher**	▲ **Higher; fewer conversions to splenectomy**
Oncological	= Non-inferior R0 and lymph node yield	▲ **0% vs 2% R1; more lymph nodes**
Operative time	= Conflicting	▲ **≈30 min longer (docking)**

■ favours RDP ■ no clear difference ■ favours comparator

05 Cost-effectiveness

Operative cost **RDP higher**

Overall cost **RDP ≤ LDP < ODP**

Quality of life **RDP > LDP**

- Theatre costs offset by shorter stay, fewer conversions and lower morbidity; costs fall with volume.
- More QALYs than LDP at similar cost (ROBOCOSTES).

06 Challenges

Operative time Docking adds time; falls with the learning curve

Team & training Dedicated robotic team; few training centres

Access High upfront cost; robots mostly in advanced economies

CONCLUSION

RDP offers greater precision, better clinical outcomes and more spleen preservation than open and laparoscopic surgery, and is cost-effective. Wider adoption depends on training and equitable access.